

BRIGHAM CITY 28 WNW, UT, USA

WMO: 725770

Lat: 41.6163N

Lon: 112.5437W

Elev: 4951

StdP: 12.25

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 04138

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.7	7.7	-2.9	5.6	8.4	1.5	7.1	10.7	14.9	34.5	12.3	32.9	1.7	N/A	0.508

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	28.2	93.6	58.4	91.4	57.7	89.1	57.1	62.2	81.4	61.1	82.1	59.9	81.9	6.6	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.4	81.5	67.8	53.6	73.6	67.6	51.2	67.1	67.8	30.4	81.1	29.5	82.4	28.6	82.2	65.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.8	10.9	9.4	-3.3	97.2	5.6	1.5	-7.3	98.3	-10.6	99.2	-13.7	100.0	-17.8	101.1		
			-3.5	64.5	5.6	1.2	-7.5	65.4	-10.8	66.1	-14.0	66.8	-18.0	67.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.9	23.5	29.1	38.7	44.8	53.8	65.4	74.9	72.3	62.7	47.7	36.3	24.8
	DBStd	18.92	8.61	8.24	7.56	7.35	7.54	8.14	5.04	5.24	7.50	8.18	8.15	8.41
	HDD50	3345	822	587	358	189	46	3	0	0	9	137	413	781
	HDD65	6945	1287	1007	816	605	353	94	4	12	122	537	862	1246
	CDD50	2589	0	0	7	34	163	465	771	692	390	66	1	0
	CDD65	714	0	0	0	0	6	106	310	238	53	1	0	0
	CDH74	10145	0	0	0	9	145	1674	4229	3165	893	30	0	0
	CDH80	4363	0	0	0	0	19	643	2087	1395	215	4	0	0
	WSAvg	4.0	2.8	3.7	4.3	4.9	4.6	4.7	4.6	4.5	4.0	3.6	3.3	2.9
	PrecAvg	13.9	1.3	1.1	1.2	1.5	1.9	1.1	0.6	0.7	0.9	1.3	1.0	1.3
Precipitation	PrecMax	22.2	3.4	2.8	3.2	3.5	4.9	3.6	1.9	2.1	3.3	4.9	2.6	3.3
	PrecMin	9.0	0.1	0.0	0.3	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.2	0.0
	PrecStd	3.6	0.9	0.7	0.7	0.8	1.4	0.9	0.5	0.5	0.7	1.0	0.6	0.8
	PrecStd	3.6	0.9	0.7	0.7	0.8	1.4	0.9	0.5	0.5	0.7	1.0	0.6	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	54.4	66.1	74.5	83.1	95.0	96.7	94.4	89.3	78.0	61.4	50.7
		MCWB	36.9	41.6	46.6	50.9	52.8	59.7	58.7	58.3	57.8	54.4	47.3	41.9
	2%	DB	41.6	49.2	61.8	69.2	78.0	90.7	94.5	92.1	85.3	71.0	57.6	44.4
		MCWB	35.6	39.4	45.1	47.7	52.7	56.2	58.4	57.8	55.6	51.6	44.3	37.8
	5%	DB	37.9	45.2	57.3	65.1	74.4	86.9	92.6	90.0	82.2	66.6	53.9	39.5
		MCWB	33.3	37.4	42.9	46.7	51.2	55.1	58.2	57.1	54.4	49.4	42.7	35.1
	10%	DB	35.1	41.5	52.9	60.7	70.0	83.6	90.5	87.7	79.6	62.6	49.6	36.3
		MCWB	31.5	35.7	40.9	45.0	50.2	54.5	57.9	56.6	54.0	47.8	40.5	32.7
	0.4%	WB	38.7	43.1	48.1	52.7	56.7	61.3	64.0	63.1	62.8	55.5	48.4	42.7
		MCDB	43.1	53.1	63.0	69.9	73.4	91.2	81.6	79.1	74.7	70.6	58.8	49.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.5	40.9	46.1	50.0	53.9	59.4	62.5	61.7	59.9	53.5	45.2	39.6
		MCDB	40.8	49.1	60.5	66.0	72.4	84.4	81.8	81.0	76.7	67.0	55.7	44.8
	5%	WB	34.2	38.9	44.2	47.8	52.5	57.6	61.5	60.3	57.8	51.7	42.8	36.1
		MCDB	37.9	45.3	57.2	62.3	70.5	81.5	82.1	81.1	75.3	63.9	51.6	39.8
	10%	WB	32.2	36.9	42.1	46.0	51.2	56.1	60.5	59.2	56.1	49.6	40.7	33.5
		MCDB	35.1	42.4	52.9	59.1	68.2	79.6	82.5	80.9	74.1	60.2	48.5	36.5
Mean Daily Temperature Range	5% DB	MDBR	15.4	15.4	19.0	21.8	22.6	27.4	28.2	27.9	26.5	22.6	19.4	14.7
		MCDBR	16.3	17.8	25.0	29.3	28.5	30.9	30.5	30.1	30.1	26.7	23.4	16.9
		MCWBR	11.5	10.9	12.9	13.8	11.9	11.3	11.4	11.8	12.6	13.1	13.2	11.2
	5% WB	MCDBR	14.2	16.5	24.0	25.5	25.4	29.2	26.6	26.5	24.8	23.1	21.9	15.4
		MCWBR	10.7	10.4	12.7	12.9	11.6	11.5	10.6	10.1	10.6	11.6	13.1	11.0
Clear-Sky Solar Irradiance	taub		0.241	0.253	0.265	0.294	0.308	0.306	0.330	0.325	0.299	0.271	0.251	0.240
	taud		2.530	2.590	2.602	2.500	2.495	2.534	2.472	2.475	2.542	2.616	2.623	2.563
	Ebn at Noon		299	311	317	311	306	306	297	296	299	299	294	291
	Edh at Noon		22	24	27	32	34	32	34	33	29	23	20	19
All-Sky Solar Radiation	RadAvg		558	848	1321	1758	2083	2440	2383	2083	1658	1140	733	471
	RadStd		80	109	94	104	157	154	91	91	100	66	62	70

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page